

Conditional Probability

Question 1. Of all failures of a certain type of computer hard drive, it is determined that in 20% of them only the sector containing the file allocation table is damaged, in 70% of them only nonessential sectors are damaged, and in 10% of the cases both the allocation sector and one or more nonessential sectors are damaged. A failed drive is selected at random and examined.

- a. What is the probability that the allocation sector is damaged?
- b. What is the probability that a nonessential sector is damaged?
- c. If the drive is found to have a damaged allocation sector, what is the probability that some nonessential sectors are damaged as well?
- d. If the drive is found to have a damaged nonessential sector, what is the probability that the allocation sector is damaged as well?
- e. If the drive is found to have a damaged allocation sector, what is the probability that no nonessential sectors are damaged?
- f. If the drive is found to have a damaged nonessential sector, what is the probability that the allocation sector is not damaged?

Solution: Let us define two events:

A = Event of the allocation sector is damaged.

N = Event of the nonessential sectors are damaged.

We are given,

Probability that only the allocation sector is damaged: $P(A \cap N^c) = 0.20$

Probability that only nonessential sectors are damaged: $P(A^c \cap N) = 0.70$

Probability that both the allocation sector and nonessential sectors are damaged: $P(A \cap N) = 0.10$

- a. The probability that the allocation sector is damaged:

$$P(A) = P(A \cap N^c) + P(A \cap N) = 0.20 + 0.10 = 0.30$$

Interpretation: There is a 30% chance that the allocation sector is damaged in a failed hard drive.

- b. The probability that a nonessential sector is damaged:

$$P(N) = P(A^c \cap N) + P(A \cap N) = 0.70 + 0.10 = 0.80$$

Interpretation: There is an 80% chance that a nonessential sector is damaged in a failed hard drive.

- c. If the drive is found to have a damaged allocation sector, the probability that some nonessential sectors are damaged as well:

$$P(N|A) = \frac{P(A \cap N)}{P(A)} = \frac{0.10}{0.30} = \frac{1}{3}$$

Interpretation: Given that the allocation sector is damaged, there is approximately a 33.3% chance that some nonessential sectors are also damaged.

- d. If the drive is found to have a damaged nonessential sector, the probability that the allocation sector is damaged as well:

$$P(A|N) = \frac{P(A \cap N)}{P(N)} = \frac{0.10}{0.80} = \frac{1}{8}$$

Interpretation: Given that a nonessential sector is damaged, there is a 12.5% chance that the allocation sector is also damaged.

- e. If the drive is found to have a damaged allocation sector, the probability that no nonessential sectors are damaged:

$$P(N^c|A) = \frac{P(A \cap N^c)}{P(A)} = \frac{0.20}{0.30} = \frac{2}{3}$$

Interpretation: Given that the allocation sector is damaged, there is approximately a 66.7% chance that no nonessential sectors are damaged.

- f. If the drive is found to have a damaged nonessential sector, the probability that the allocation sector is not damaged:

$$P(A^c|N) = \frac{P(A^c \cap N)}{P(N)} = \frac{0.70}{0.80} = \frac{7}{8}$$

Interpretation: Given that a nonessential sector is damaged, there is an 87.5% chance that the allocation sector is not damaged.

Question 2. During the monsoon season in a coastal city, there is a 60% chance of rain on any given day. When it rains, there is a 50% chance that strong winds will accompany the rain. On days when it does not rain, there is still a 20% chance that the city will experience strong winds due to nearby storms.

- What is the probability that it will not rain on a random day during the monsoon season?
- What is the probability that strong winds will not occur when it does not rain?
- What is the probability that, on a random day of the monsoon season, it will rain and strong wind?
- What is the probability that it will not rain and there will be no strong winds on a random day?
- Suppose, there is a 38% chance that there will be strong winds on a random day regardless of whether it rains or not. Determine the probability that it rained that day if strong winds are observed.

Solution: Let,

R = Event of rain on a random day

W = Event of wind on a random day

Here, $P(R) = 0.6$; $P(W|R) = 0.5$; $P(W|R') = 0.2$

- a. Probability that it will not rain on a random day during the monsoon season,

$$P(R') = 1 - P(R) = 0.4$$

Comment: 40% chance that it will not rain on a random day during the monsoon season.

- b. The probability that strong winds will not occur given that it does not rain,

$$P(W'|R') = 1 - P(W|R') = 1 - 0.2 = 0.8$$

Comment: 80% chance that strong winds will not occur when it does not rain.

- c. The probability that, it will rain and strong wind,

$$P(R \cap W) = P(W|R) \times P(R) = 0.5 \times 0.6 = 0.3$$

Comment: 30% chance that it will rain and strong wind.

- d. The probability that it will not rain and there will be no strong winds on a random day,

$$P(R' \cap W') = P(W'|R') \times P(R')$$

Here,

$$P(R') = 1 - P(R) = 0.4$$

$$P(W'|R') = 1 - P(W|R') = 1 - 0.2 = 0.8$$

$$\therefore P(R' \cap W') = P(W'|R') \times P(R') = 0.4 \times 0.8 = 0.32$$

Comment: 32% chance that it will not rain and not any strong winds.

- e. Here, Probability that there will be strong winds on a random day regardless of whether it rains or not, $P(W) = 0.38$. The probability that it rained that day if strong winds are observed,

$$P(R|W) = \frac{P(R \cap W)}{P(W)} = \frac{0.3}{0.38} = 0.79$$

Comment: Approximately, 79% chance that it rained that day if strong winds are observed.

Question 3. At BRAC University, a survey was conducted among the students of the Department of Computer Science and Engineering. The survey revealed the following statistics: 40% of the students reported difficulties in understanding Algorithms. 30% of the students reported difficulties in understanding Data Structures. 20% of the students reported difficulties in both Algorithms and Data Structures. A student is selected at random from the department.

- a. If the student reported difficulties in Data Structures, estimate the probability that they also reported difficulties in Algorithms.
- b. If the student reported difficulties in Algorithms, estimate the probability that they also reported difficulties in Data Structures
- c. Calculate the probability that the student reported difficulties in either Algorithms or Data Structures.

- d. Given that a student did not report difficulties in Algorithms, estimate the probability that they reported difficulties in Data Structures.
- e. What is the probability that the student did not report difficulties in both Algorithms and Data Structures?

Solution: Given,

$P(A) = 0.40$ (Probability of difficulties in Algorithms)

$P(B) = 0.30$ (Probability of difficulties in Data Structures)

$P(A \cap B) = 0.20$ (Probability of difficulties in both Algorithms and Data Structures)

- a. The conditional probability $P(A | B)$ is calculated as:

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.20}{0.30} = \frac{2}{3} \approx 0.6667$$

- b. The conditional probability $P(B | A)$ is calculated as:

$$P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{0.20}{0.40} = \frac{1}{2} = 0.50$$

- c. The probability $P(A \cup B)$ is calculated as:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.40 + 0.30 - 0.20 = 0.50$$

- d. Let A^c denote the complement of A , the event that the student did not report difficulties in Algorithms.

$$P(A^c) = 1 - P(A) = 1 - 0.40 = 0.60$$

Using the law of total probability:

$$P(B) = P(B | A)P(A) + P(B | A^c)P(A^c)$$

Substituting the known values:

$$0.30 = (0.50 \times 0.40) + P(B | A^c) \times 0.60$$

$$0.30 = 0.20 + 0.60P(B | A^c)$$

$$0.10 = 0.60P(B | A^c)$$

$$P(B | A^c) = \frac{0.10}{0.60} = \frac{1}{6} \approx 0.1667$$

- e. The probability $P(A^c \cap B^c)$ is calculated as:

$$P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - 0.50 = 0.50$$

Question 4. A survey conducted by an e-commerce company wants to know people's buying preference whether online platforms or traditional platforms among both male and female participants. It is found that 35% of total participants prefer online platforms to purchase. Where, 70% of Participants who prefer online platform were female and 60% of participants who didn't prefer online platform were male. A participant randomly selected

- What is the probability that the participant prefers traditional platforms?
- What is the probability that the participant is female and prefers online platforms?
- What is the probability that the participant is male and prefers online platforms?
- What is the probability that the participant is female and prefers traditional platforms?
- If the probability of being a woman is 55 %, what is the probability that she will prefer online platforms?

Solution: Given,

The probability of preferring online platforms is: $P(O) = 0.35$

The probability of being women given that participant prefer online platforms is: $P(F|O) = 0.7$

The probability of being men given that participant didn't prefer online platforms is: $P(M|O') = 0.6$

- If, O' be the event of traditional platform (or, not preferring the online platform), then the probability that the participant prefers traditional platforms,

$$P(O') = 1 - P(O) = 0.65$$

- Probability that the participant is female and prefers online platforms,

$$P(F \cap O) = P(F|O) \times P(O) = 0.7 \times 0.35 = 0.245$$

- The probability of being men given that participant prefer online platforms is:

$$P(M|O) = 1 - P(F|O) = 0.3$$

Probability that the participant is male and prefers online platforms,

$$P(M \cap O) = P(M|O) \times P(O) = 0.3 \times 0.35 = 0.105$$

- The probability of being female given that participant prefer traditional platforms is,

$$P(F|O') = 1 - P(M|O') = 0.4$$

Probability that the participant is female and prefers traditional platforms,

$$P(F \cap O') = P(F|O') \times P(O') = 0.4 \times 0.65 = 0.26$$

- Probability that participant will prefer online platforms given that participant is female,

$$P(O|F) = \frac{P(O \cap F)}{P(F)} = \frac{0.245}{0.55} = 0.445$$

Question 5. In a cultural society of BRACU of 100 students, 70 are girls and 30 are boys. Among the girls, 20 are in the science club, and among the boys, 10 are in the science club.

- What is the probability that a randomly selected student is a girl?
- What is the probability that a randomly selected student is in the science club?
- What is the probability that a student is a girl given that the student is in the science club?
- What is the probability that a student is in the science club given that the student is a boy?

Solution:

- Probability that the student is a girl:

$$P(G) = \frac{70}{100} = 0.7$$

- Probability that the student is in the science club:

Total in the science club = 20 (girls) + 10 (boys) = 30.

$$P(SC) = \frac{30}{100} = 0.3$$

- Probability that a student is a girl given that the student is in the science club:

$$P(G|SC) = \frac{P(G \cap SC)}{P(SC)} = \frac{20/100}{30/100} = \frac{20}{30} = \frac{2}{3} \approx 0.67$$

- Probability that a student is in the science club given that the student is a boy:

$$P(SC|B) = \frac{P(SC \cap B)}{P(B)} = \frac{10/100}{30/100} = \frac{10}{30} = \frac{1}{3} \approx 0.33$$

Question 6. The probability that a married man watches a certain TV show is 0.4 and that a married woman watches the show is 0.5. The probability that a man watches the show, given that his wife does, is 0.7.

- Find the probability that a married couple watches the show.
- Find the probability that a wife watches the show given that her husband does.
- Find the probability that at least one person of a married couple will watch the show.

Solution: Let us define two events H and W as follows:

H: Husband watches the show

W: Wife watches the show

It is given that, $P(H) = 0.4$, $P(W) = 0.5$ and $P(H|W) = 0.7$

- $$P(H \cap W) = P(H|W) \times P(W)$$

$$= 0.7 \times 0.5$$

$$= 0.35$$

- $$P(W|H) = \frac{P(H \cap W)}{P(H)} = \frac{0.35}{0.4} = 0.875$$

$$\begin{aligned}
 c. \quad P(H \cup W) &= P(H) + P(W) - P(H \cap W) \\
 &= 0.4 + 0.5 - 0.35 \\
 &= 0.55
 \end{aligned}$$

Question 7. The Roads and Highways Department (RHD) conducted a survey during Christmas vacation. The probability that a vehicle entering a hillside tunnel has license plates is 0.33 and the probability that it is a campervan is 0.48. The probability that it is a campervan with license plates is 0.09.

- What is the probability that a campervan entering the tunnel has license plates?
- What is the probability that a vehicle with license plates entering the tunnel is a camper van?
- What is the probability that a vehicle entering the tunnel does not have license plates or is not a campervan?

Solution: Let,

C = the vehicle is a campervan

L = the vehicle has license plates

Given that, $P(L) = 0.33$, $P(C) = 0.48$, $P(L \cap C) = 0.09$

- The probability that a campervan entering the tunnel has license plates:

$$P(L|C) = \frac{P(L \cap C)}{P(C)} = \frac{0.09}{0.48} = 0.1875$$

- The probability that a vehicle with license plates entering the tunnel is a camper van:

$$P(C|L) = \frac{P(L \cap C)}{P(L)} = \frac{0.09}{0.33} = 0.27$$

- The probability that a vehicle entering the tunnel does not have license plates or is not a campervan:

$$P(L' \cup C') = 1 - P(L \cap C) = 1 - 0.09 = 0.91$$

Question 8. In a city library, a survey was conducted among library visitors. The survey provided the following data: 50% of the visitors reported difficulties in finding books in the Fiction section. 40% of the visitors reported difficulties in finding books in the Non-Fiction section. 25% of the visitors reported difficulties in both the Fiction and Non-Fiction sections. 15% of the visitors reported difficulties in neither the Fiction nor the Non-Fiction sections. A visitor is selected at random from the library.

- If the visitor reported difficulties in finding books in the Non-Fiction section, estimate the probability that they also reported difficulties in finding books in the Fiction section.

- b. If the visitor reported difficulties in finding books in the Fiction section, estimate the probability that they also reported difficulties in finding books in the Non-Fiction section.
- c. Calculate the probability that the visitor reported difficulties in either the Fiction or the Non-Fiction section.
- d. Given that a visitor did not report difficulties in finding books in the Non-Fiction section, estimate the probability that they reported difficulties in finding books in the Fiction section.
- e. What is the probability that the visitor did not report difficulties in either the Fiction or the Non-Fiction sections?

Solution: Let's denote the following events:

F: The event that a visitor reported difficulties in finding books in the Fiction section.

N: The event that a visitor reported difficulties in finding books in the Non-Fiction section.

We are given: $P(F) = 0.50$; $P(N) = 0.40$; $P(F \cap N) = 0.25$; $P(F^c \cap N^c) = 0.15$

- a. The probability that a visitor who reported difficulties in the Non-Fiction section also reported difficulties in the Fiction section is:

$$P(F|N) = \frac{P(F \cap N)}{P(N)} = \frac{0.25}{0.40} = 0.625$$

- b. The probability that a visitor who reported difficulties in the Fiction section also reported difficulties in the Non-Fiction section is:

$$P(N|F) = \frac{P(F \cap N)}{P(F)} = \frac{0.25}{0.50} = 0.50$$

- c. To find the probability that a visitor reported difficulties in either the Fiction or the Non-Fiction section:

$$\begin{aligned} P(F \cup N) &= P(F) + P(N) - P(F \cap N) \\ P(F \cup N) &= 0.50 + 0.40 - 0.25 = 0.65 \end{aligned}$$

- d. To find the probability that a visitor who did not report difficulties in the Fiction section reported difficulties in the Non-Fiction section:

$$\begin{aligned} P(F^c) &= 1 - P(F) = 1 - 0.50 = 0.50 \\ P(N \cap F^c) &= P(N) - P(F \cap N) = 0.40 - 0.25 = 0.15 \end{aligned}$$

$$P(N|F^c) = \frac{P(N \cap F^c)}{P(F^c)} = \frac{0.15}{0.50} = 0.30$$

- e. To find the probability that the visitor did not report difficulties in either the Fiction or the Non-Fiction section:

$$P(F^c \cap N^c) = 0.15$$

So, the probability that the visitor did not report difficulties in either the Fiction or the Non-Fiction section is 0.15.

Bayes Theorem

Question 1. A certain delivery service offers both express and standard delivery. Seventy-five percent of parcels are sent by standard delivery, and 25% are sent by express. Of those sent standard, 80% arrive the next day, and of those sent express, 95% arrive the next day. A record of a parcel delivery is chosen at random from the company's files.

- a. What is the probability that the parcel was shipped express and arrived the next day?
- b. What is the probability that it arrived the next day?
- c. Given that the package arrived the next day, what is the probability that it was sent express?

Solution: Let us define the following events:

S = the parcel is sent by standard delivery.

E = the parcel is sent by express delivery.

A = the parcel arrives the next day.

Given, Probability that a parcel is sent by standard delivery: $P(S) = 0.75$

Probability that a parcel is sent by express delivery: $P(E) = 0.25$

Probability that a parcel sent by standard delivery arrives the next day: $P(A | S) = 0.80$

Probability that a parcel sent by express delivery arrives the next day: $P(A | E) = 0.95$

- a. The probability that the parcel was shipped express and arrived the next day:

This is the joint probability of occurring both events (E) and (A):

$$P(E \cap A) = P(E) \times P(A | E) = 0.25 \times 0.95 = 0.2375$$

Interpretation: There is a 23.75% chance that a parcel was shipped express and arrived the next day.

- b. The probability that it arrived the next day: This is the total probability of happening (A), considering both standard and express deliveries:

$$P(A) = P(S) \times P(A | S) + P(E) \times P(A | E)$$

$$= 0.75 \times 0.80 + 0.25 \times 0.95$$

$$= 0.8375$$

Interpretation: There is an 83.75% chance that a randomly chosen parcel from the company's files arrived the next day.

- c. Given that the package arrived the next day, the probability that it was sent express:

$$P(E | A) = \frac{P(E \cap A)}{P(A)} = \frac{0.2375}{0.8375} = 0.2836$$

Interpretation: Given that a parcel arrived the next day, there is approximately a 28.36% chance that it was sent by express delivery.

Question 2. A certain level of customer dissatisfaction is present in about 1 out of 1000 customers in a given population. Suppose there is a customer feedback survey, and customers are asked to rate their satisfaction level. The survey results show that a positive rating is given with a probability of 10% for a dissatisfied customer and with a probability of 95% for a satisfied customer.

- If a customer is selected at random from this population, what is the probability that the feedback survey will result in a positive rating? Interpret your findings.
- If the feedback survey of a selected customer results in a negative rating, what is the probability that they are actually dissatisfied?

Solution: Let,

D = Event of dissatisfied customer

S = Event of satisfied customer

PR = Event of positive rating

NR = Event of negative rating

- The probability that the feedback survey will result in a positive rating,

$$P(PR) = P(PR|D) \times P(D) + P(PR|S) \times P(S)$$

$$\text{Here, } P(D) = \frac{1}{1000} = 0.001; P(S) = 1 - P(D) = 0.999; P(PR|D) = 0.10; P(PR|S) = 0.95$$

$$\therefore P(PR) = P(PR|D) \times P(D) + P(PR|S) \times P(S) \\ = (0.10 \times 0.001) + (0.999 \times 0.95) = 0.95$$

- The probability that they are actually dissatisfied when selected customer results in a negative rating, $P(D|NR) = (P(NR|D) \times P(D))/P(NR)$

$$\therefore P(NR) = 1 - P(PR) = 1 - 0.95 = 0.05$$

$$\therefore P(NR|D) = 1 - P(PR|D) = 1 - 0.10 = 0.90$$

$$\therefore P(D) = 0.001$$

$$\therefore P(D|NR) = (P(NR|D) \times P(D))/P(NR) = ((0.90 \times 0.001))/0.05 = 0.018$$

Question 3. A company has developed a new drug test that can detect a particular substance in a person's blood. The test has the following characteristics: **True Positive Rate (Sensitivity):** 97% (the probability that the test correctly identifies a person with the substance as positive). **False Positive Rate:** 5% (the probability that the test incorrectly identifies a person without the substance as positive). It is known that 1% of the population uses this substance. A person is randomly selected from the population and tests positive. **What is the probability that this person actually uses the substance?**

Solution: Let's define the events:

U : The person uses the substance.

U' : The person does not use the substance.

T^+ : The test result is positive.

We want to find the probability that a person actually uses the substance given a positive test result, $P(U | T^+)$. By **Bayes' Theorem**:

$$P(U | T^+) = \frac{P(T^+ | U) \cdot P(U)}{P(T^+)}$$

Step 1: Find Each Component for Bayes' Theorem

- **Prior Probability:** The probability that a randomly selected person uses the substance:

$$P(U) = 0.01$$

- **Complement Probability:** The probability that a randomly selected person does not use the substance:

$$P(U') = 1 - P(U) = 0.99$$

- **True Positive Rate (Sensitivity):** The probability that the test is positive given the person uses the substance:

$$P(T^+ | U) = 0.97$$

- **False Positive Rate:** The probability that the test is positive given the person does not use the substance:

$$P(T^+ | U') = 0.05$$

- **Total Probability of Testing Positive (Law of Total Probability):**

$$P(T^+) = P(T^+ | U) \cdot P(U) + P(T^+ | U') \cdot P(U')$$

Substitute the values:

$$P(T^+) = (0.97 \times 0.01) + (0.05 \times 0.99)$$

$$P(T^+) = 0.0097 + 0.0495 = 0.0592$$

Step 2: Apply Bayes' Theorem

Now substitute these values into Bayes' Theorem:

$$P(U | T^+) = \frac{P(T^+ | U) \cdot P(U)}{P(T^+)}$$

$$P(U | T^+) = \frac{0.97 \times 0.01}{0.0592}$$

$$P(U | T^+) = \frac{0.0097}{0.0592} \approx 0.1638$$

Final Answer: The probability that a person who tests positive actually uses the substance is approximately **16.38%**.

Interpretation: Even though the test has a high sensitivity (97%) and a relatively low false positive rate (5%), the probability that a person who tests positive is actually using the substance is still quite low (16.38%). This is due to the low prevalence (1%) of substance use in the population, demonstrating the significance of considering base rates in diagnostic testing scenarios.

Question 4. During the recent student protests in Bangladesh, a survey was conducted among the students of BRAC University. The survey revealed that: 25% of the students physically participated in the protests. Among those who physically participated in the protests, 60% were active in organizing online campaigns. Among those who did not physically participate in the protests, 80% were also active in organizing online campaigns.

- What is the prior probability that a randomly selected student physically participated in the protests?
- Given that a student physically participated in the protests, what is the probability that they were active in organizing online campaigns?
- If a student was active in organizing online campaigns, what is the probability that they actually physically participated in the protests?

Solution: Let,

$P(P)$ be the probability that a student physically participated in the protests.

$P(O | P)$ be the probability that a student was active in organizing online campaigns given they participated in the protests.

$P(O | P^c)$ be the probability that a student was active in organizing online campaigns given they did not participate in the protests.

$P(O)$ be the probability that a student was active in organizing online campaigns.

$P(P | O)$ be the probability that a student physically participated in the protests given they were active in organizing online campaigns.

- a. The prior probability that a randomly selected student physically participated in the protests is given as:

$$P(P) = 0.25$$

- b. The probability that a student who physically participated in the protests was active in organizing online campaigns is:

$$P(O | P) = 0.60$$

- c. To find the probability that a student physically participated in the protests given they were active in organizing online campaigns, we use Bayes' theorem:

$$P(P | O) = \frac{P(O | P) \cdot P(P)}{P(O)}$$

First, calculate $P(O)$ using the law of total probability:

$$P(O) = P(O | P) \cdot P(P) + P(O | P^c) \cdot P(P^c)$$

$$P(P^c) = 1 - P(P) = 1 - 0.25 = 0.75$$

$$P(O) = (0.60 \cdot 0.25) + (0.80 \cdot 0.75)$$

$$P(O) = 0.15 + 0.60 = 0.75$$

Now, calculate $P(P | O)$:

$$P(P | O) = \frac{P(O | P) \cdot P(P)}{P(O)}$$

$$P(P | O) = \frac{0.60 \cdot 0.25}{0.75} = \frac{0.15}{0.75} = 0.20$$

Thus, the probability that a student physically participated in the protests given they were active in organizing online campaigns is 0.20 or 20%.

Question 5. Two factories supply light bulbs to the market. Bulbs from factory X work for over 5000 hours in 99% of cases, whereas bulbs from factory Y work for over 5000 hours in 95% of cases. It is known that factory X supplies 60% of the total bulbs available in the market.

- What is the probability that a purchased bulb will work for longer than 5000 hours?
- Given that a light bulb works for more than 5000 hours, what is the probability that it was supplied by factory Y?
- Given that a light bulb work does not work for more than 5000 hours, what is the probability that it was supplied by factory X?

Solution: Let, H = bulb works over 5000 hours; X = A bulb comes from factory X; Y = A bulb comes from factory Y

a. $P(H) = P(H|X)P(X) + P(H|Y)P(Y) = (0.99)(0.6) + (0.95)(0.4) = 0.974$

b. $P(Y|H) = \frac{P(H|Y)P(Y)}{P(H)} = \frac{0.95 \times 0.4}{0.974} = 0.39$

$$c. P(X|H^c) = \frac{P(H^c|X)P(X)}{P(H^c)} = \frac{P(H^c|X)P(X)}{1 - P(H)} = \frac{(1 - 0.99)(0.6)}{1 - 0.974} = 0.23$$

Question 6. A research team working on finding a treatment for a newly transmitted virus. Where, 60% of patients were given medicine and a placebo was given to others. From the treatment group 70% and from the placebo group 30% patient recovered within 1 month.

- What is the probability that placebo was given and the patient will be recovered?
- What is the probability that a patient recovered from it?
- Given that a patient recovered, what is the probability that patient belong to medicine group?

Solution: Given that,

The probability that medicine given to a patient, $P(M) = 0.6$

The probability that placebo given to a patient, $P(P) = 0.4$

The probability that medicine given to a patient and the patient recovered, $P(R|M) = 0.7$

The probability that placebo given to a patient and the patient recovered, $P(R|P) = 0.3$

- The probability that placebo was given and the patient will be recovered,

$$P(R \cap P) = P(R|P) \times P(P) = 0.3 \times 0.4 = 0.12$$

- The probability that a patient recovered from it,

$$P(R) = P(R|M) \times P(M) + P(R|P) \times P(P) = 0.7 \times 0.6 + 0.3 \times 0.4 = 0.54$$

- Given that patient recovered, the probability that patient belong to medicine group,

$$P(M|R) = \frac{P(M \cap R)}{P(R)} = P(R|M) \times \frac{P(M)}{P(R)} = \frac{0.7 \times 0.6}{0.54} = 0.78$$

Question 7. A blood test gives 96% positive (+) results for diseased people and 3% positive results for healthy people. In a community 2% people have diseases. A person is randomly selected from the community and gives the test.

- What is the probability that the test will be positive?
- If the test is positive, what is the probability that the person has the disease?
- If the test is positive, what is the probability that the person has not the disease?

Solution: Let us define, D for disease; H for healthy; + for positive test result.

It is given that, $P(+|D) = 0.96$, $P(+|H) = 0.03$, $P(D) = 0.02$, $P(H) = 0.98$

- $$P(+) = P(D \cap +) + P(H \cap +) = P(D).P(+|D) + P(H).P(+|H)$$

$$= 0.02 \times 0.96 + 0.98 \times 0.03$$

$$= 0.0486$$

- b. $P(D|+) = \frac{P(D \cap +)}{P(+)} = \frac{P(D) \times P(+|D)}{P(+)} = \frac{0.02 \times 0.96}{0.0486} = 0.3951$
- c. $P(H|+) = 1 - P(D|+) = 1 - 0.3951 = 0.6049$

Question 8. A lie-detector machine has the property that 90% of the guilty suspects are properly judged while 10% of the guilty suspects are improperly found innocent. On the other hand, innocent suspects are misjudged 1% of the time. Suppose, only 5% suspects have ever committed a crime.

- What is the probability that the suspect is innocent?
- What is the probability that the suspect is judged guilty?
- If the machine indicates that a selected suspect is guilty, what is the probability that he is innocent?

Solution: Here the defined events, G=Suspect is guilty; I=Suspect is innocent; JG= Suspect is judged guilty; JI=Suspect is judged innocent

Given that, $P(JG|G) = 0.90$, $P(JI|G) = 0.10$, $P(JG|I) = 0.01$, $P(G) = 0.05$

- $P(I) = 1 - P(G) = 1 - 0.05 = 0.95$
- $P(JG) = P(JG|I)P(I) + P(JG|G)P(G) = 0.01 \times 0.95 + 0.90 \times 0.05 = 0.0545$
- $P(I|JG) = \frac{P(JG|I)P(I)}{P(JG|I)P(I) + P(JG|G)P(G)} = \frac{0.01 \times 0.95}{0.01 \times 0.95 + 0.90 \times 0.05} = 0.1743$

Question 9.

In a bustling city, there are two popular types of local cafes: artisanal coffee shops and casual diners. The following statistics are available from a recent citywide survey: 70% of the cafes are artisanal coffee shops, while 30% are casual diners. 10% of the artisanal coffee shops offer live music performances. 25% of the casual diners offer live music performances. A city guidebook randomly selects a cafe known for its live music performances.

- What is the probability that a randomly selected cafe with live music is an artisanal coffee shop?
- Given that a cafe is known to be an artisanal coffee shop, what is the probability that it offers live music performances?
- What is the probability that a randomly selected cafe is a casual diner given that it offers live music performances?

Solution:

Let's denote the following events:

A: The event that a cafe is an artisanal coffee shop.

C: The event that a cafe is a casual diner.

M: The event that a cafe offers live music performances.

We are given: $P(A) = 0.70$; $P(C) = 0.30$; $P(M | A) = 0.10$ (probability of offering live music given that the cafe is an artisanal coffee shop); $P(M | C) = 0.25$ (probability of offering live music given that the cafe is a casual diner)

- a. Probability that a Cafe with Live Music is an Artisanal Coffee Shop:

Using Bayes' Theorem:

$$P(A|M) = \frac{P(M|A) \cdot P(A)}{P(M)}$$

First, we need to find $P(M)$, the total probability of a cafe offering live music:

$$P(M) = P(A) \cdot P(M|A) + P(C) \cdot P(M|C) = 0.145$$

compute $P(A|M)$:

$$P(A|M) = \frac{0.10 \times 0.70}{0.145} \approx 0.483$$

So, the probability that a cafe with live music is an artisanal coffee shop is approximately 0.483.

- b. Probability that an Artisanal Coffee Shop Offers Live Music:

This is directly given by:

$$P(M | A) = 0.10$$

So, the probability that an artisanal coffee shop offers live music is 0.10.

- c. Probability that a Cafe is a Casual Diner Given that it Offers Live Music:

Using Bayes' Theorem:

$$P(C|M) = \frac{P(M|C) \cdot P(C)}{P(M)} = 0.517$$

So, the probability that a cafe with live music is a casual diner is approximately 0.517.

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